AJAYKUMAR J

AI & Data Science | Data Analytics | Machine Learning | IoT Systems

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PROFESSIONAL SUMMARY

B.Tech student in Artificial Intelligence & Data Science with hands-on experience in data analytics, machine learning model development, dashboard visualization, and IoT-based system integration. Strong foundation in data preprocessing, exploratory data analysis, and real-time application deployment. Proven ability to deliver end-to-end technical projects independently.

EDUCATION

B.Tech – Artificial Intelligence & Data Science | Ganadipathy Tulsi's Jain Engineering College, Vellore
2023 – 2027 (Pursuing) | CGPA: 8.1

PUBLICATIONS

"The Implications of Artificial Intelligence and Automation in India, Specifically with Regard to Job Displacement and Opportunities"
International Journal Research Publication Analysis (IJRPA), ISSN 2456-9995, Volume 2, Issue 8 — August 2026

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript, Dart, Embedded C/C++, NLP Basics
Data Analytics: Pandas, NumPy, Matplotlib, Data Cleaning, EDA, KPI Dashboards
Machine Learning: Scikit-learn, XGBoost, Model Evaluation, Feature Engineering
Mobile & Web: Flutter, Streamlit, HTML5, CSS3, Bootstrap
Backend & Database: Firebase Realtime Database, Firestore, REST API, WebSocket
IoT & Embedded Systems: Raspberry Pi, ESP32, ESP32-C3, IR Sensors, L298N Motor Driver, BLE HID
Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Android Studio

INTERNSHIP EXPERIENCE

Frontend Developer — *BA Technology, Chennai*

- Built PandaAtlas, a hotel booking website.

Mobile App Development — *Myme Techies, Vellore*

- Built AstraRoom, a room automation application.

PROJECTS

AquaControl – IoT Smart Aquarium Controller

- Built a full-stack Flutter + Firebase application for real-time control of an ESP32-based aquarium system, published as a signed Android APK.
- Developed ESP32 firmware (v2.5.x) with Firebase Realtime Database and WebSocket integration for bidirectional device communication.
- Implemented features including sonar radar device discovery, feed timing history, water refill logging, and offline connection guards.
- Designed light/purple themed UI with custom wave logo, resolved boot crash issues related to WiFi initialization and partition size limits.

Astra Room Automation – Smart Home App (In Development)

- Designing a Flutter-based smart home automation app targeting ESP32-C3 devices with a cyberpunk/glassmorphism UI.
- Implementing Riverpod state management with eight device control cards, scene modes, and timer scheduling.
- Integrating real-time WebSocket communication for bidirectional ESP32-C3 device control and status monitoring.

Sales Data Analysis & Business Intelligence Dashboard

- Performed end-to-end data analysis on a retail sales dataset to identify revenue trends and seasonal patterns.
- Conducted thorough data cleaning and exploratory data analysis (EDA) to ensure data integrity.
- Delivered actionable insights for sales forecasting and business optimization using KPI dashboards.

Telecom Customer Churn Prediction System

- Developed a machine learning model to predict telecom customer churn with high accuracy.
- Performed data preprocessing, feature engineering, and rigorous model validation.
- Implemented XGBoost classifier and deployed a real-time prediction application using Streamlit.

Vision Mark – AI-Powered Face Recognition Attendance & Surveillance Rover (In Development)

- Designed an AI-based rover integrating computer vision and IoT systems for autonomous surveillance.
- Implemented real-time face recognition using OpenCV for automated attendance tracking.
- Integrated ESP32-CAM module for live video streaming and remote monitoring capability.

CtrlX – ESP32 BLE HID Controller

- Built a custom ESP32-based Bluetooth HID controller with joystick, OLED display, and multiple operating modes.
- Implemented modes including Mouse, Presentation, Gaming, Media, Eye Animation, and Arcade (Snake, Pong, Flappy Bird).
- Resolved BLE library compatibility issues with BleCombo and Keyboard/Mouse API for stable wireless connectivity.

Autonomous Line Following Robot

- Built a line-following robotic vehicle using IR sensor modules for real-time path detection.
- Implemented motor control logic using L298N driver for smooth and accurate navigation.
- Optimized sensor calibration to ensure consistent performance across varied surface conditions.

Mental Health Chatbot – Emotion Detection & Support System

- Developed a rule-based NLP chatbot that detects user emotions such as stress, sadness, and happiness using text pattern analysis.
- Implemented emotion classification using Python, Regex, and Natural Language Processing techniques.
- Designed a supportive response generation system to provide empathetic replies based on detected emotional states.

LANGUAGES

English – Professional Proficiency   Tamil – Native   Spanish – Beginner

DECLARATION

I hereby declare that the information provided above is true to the best of my knowledge and belief.